

Artificial Intelligence & Machine Learning

Comprehensive Study & Practical Algorithm Guide

Subject: Artificial Intelligence & ML

Topic Coverage: Search Algorithms & Classification

Document Version: 1.0

1. Informed & Uninformed Search Algorithms

Search algorithms enable an AI agent to navigate state spaces and discover optimal action sequences reaching a designated goal state.

A* Search Evaluation Function:

$$f(n) = g(n) + h(n)$$

Where $g(n)$ is path cost from start to node n , and $h(n)$ is heuristic estimation from n to goal.

Algorithm	Type	Strategy / Cost Function	Completeness & Optimality
Uniform Cost Search (UCS)	Uninformed	Expands lowest path cost $g(n)$	Complete & Optimal (if step costs $\epsilon > 0$)
Greedy Best-First Search	Informed	Expands lowest heuristic cost $h(n)$	Not complete, Not optimal
A* Search	Informed	Combines cost & heuristic: $f(n) = g(n) + h(n)$	Optimal if $h(n)$ is admissible & consistent

2. Supervised Learning: Classification Models

Supervised learning algorithms map input feature vectors X to target categorical labels y using training dataset pairs.

K-Nearest Neighbors (K-NN)

A non-parametric, instance-based learning algorithm that classifies new data points based on majority voting among its K closest neighbors.

Distance Metrics for K-NN:

Euclidean Distance: $d(p, q) = \sqrt{(\sum_{i=1}^n (q_i - p_i)^2)}$

Manhattan Distance: $d(p, q) = \sum_{i=1}^n |q_i - p_i|$

Random Forest Classifier

An ensemble learning method that constructs a multitude of Decision Trees during training and outputs the mode of the classes predicted by individual trees.

- Bagging (Bootstrap Aggregating):** Subsamples training samples with replacement to train independent decision trees.
- Feature Subspace Randomization:** Selects a random subset of features at each split point to reduce correlation among trees.

- **Impurity Metrics:** Splitting decisions rely on Gini Impurity or Information Entropy:

$$Gini = 1 - \sum_{i=1}^C (p_i)^2 \quad | \quad Entropy = - \sum_{i=1}^C p_i \log_2(p_i)$$

3. Model Evaluation & Performance Metrics

Classification performance is evaluated using metrics derived from the Confusion Matrix (True Positives, False Positives, True Negatives, False Negatives).

Classification Metrics Formulas:

- **Accuracy:** $(TP + TN) / (TP + TN + FP + FN)$
- **Precision:** $TP / (TP + FP)$ (Focuses on minimizing False Positives)
- **Recall (Sensitivity):** $TP / (TP + FN)$ (Focuses on minimizing False Negatives)
- **F1-Score:** $2 \cdot (Precision \cdot Recall) / (Precision + Recall)$ (Harmonic Mean)